

# Draft Ideas Thesis

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May - June 2026

*Some ideas for the main problem and main objective based on meetings, papers, and research I've done. It can be modified depending on the advisor's opinions.*

**Update! Introduction is being redacted and divided into paragraphs, with the intention of following the last idea from the last paragraph to connect the medical field to the computational application.**

## 1 Lists of topics (useful)

This part is important to note because those were collected during meetings with Doctor Alejandro and Doctor Salvador, both ophthalmologic surgeons specialized in cataract surgery.

More than topics, these are separate lists that will contain problems during OP and post OP, important abilities for cataract surgery when the cataract is harder, importance of parameter selection

From Guadalajara doc:

### **Skill assessment abilities:**

1. Quickness
2. Difficulty
3. Precision: For post op recovery (faster)

**Difficult and possible situations during surgery:** (From doc Alejandro feedback on our last meeting, we cannot be super specific on certain situations during surgery)

1. Dysfunctional state of the endothelium: Risk in retinal detachment
2. Poor pupil dilatation: important during surgery

### **Importance of surgery parameters (during op):**

1. Ultrasound
2. Endothelium protection
3. Fem-to energy
4. Instruments tips
5. Quickness
6. Angulation
7. Capsulorhexis (incision)

**Various evaluation topics:**

1. Technique types
2. Hand movement
3. Pre-op parameters
4. Eye size
5. Tissue state

Also, some human body structures were discussed with doc Alejandro, but his advice was to avoid that variability.

## **2 Data available**

1. Exam ICO-OSCAR
2. Mansoor and doc Alejandro (Pre-dataset): iol calculations, before and after pic - b&w, 15 min surgery video
3. Cataract LMM dataset

**Important to note that there are some needed implementations for this project, like a **MULTI-DATA** and **MULTI-MODAL** approach, having in mind the amount of data diversity we will need (Images, numeric parameters, numeric evaluations, videos, and also those will need different processing models)**

### 3 Main ideas

From I look up on different paper platforms, there are not papers that present the idea of combining the different hardness levels of the cataracts and how every level need a different surgery performance evaluating skill assessment.

Here I did a little diagram where I think I can start to work, having in mind that I need to limit certain ideas because of our available data (Pre-dataset).

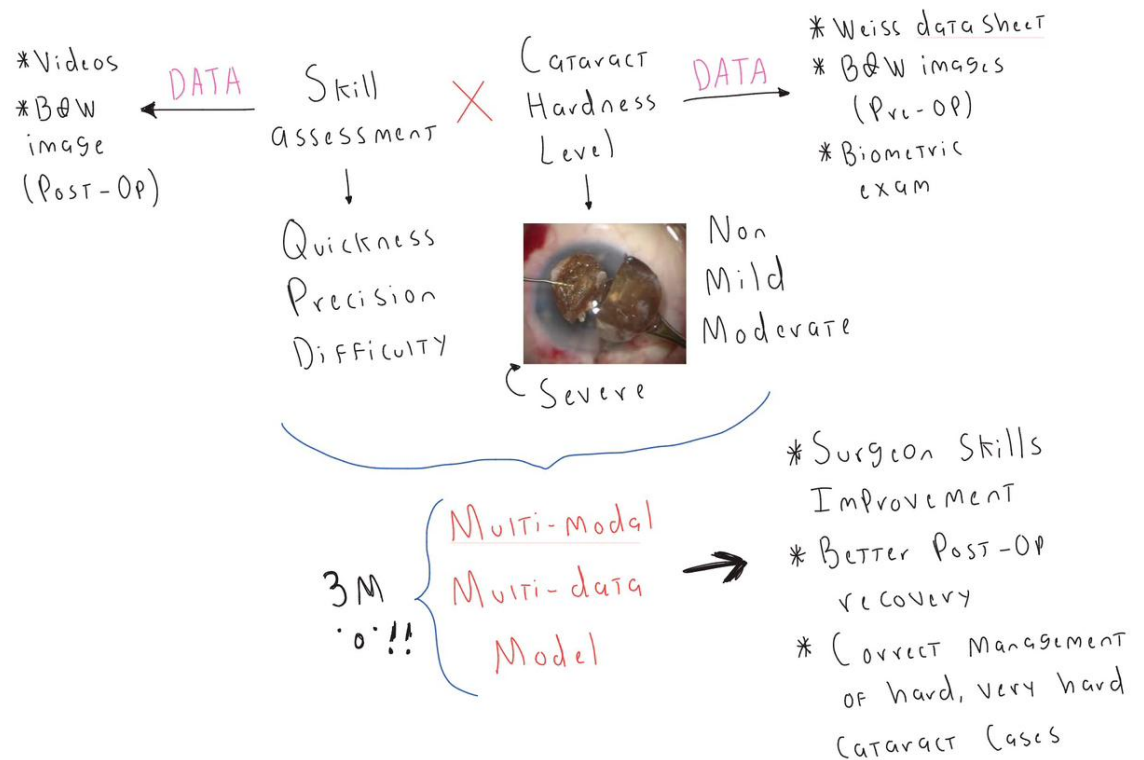


Figure 1: First diagram.

→ Create a MULTI-MODAL MULTI-DATA MODEL (main) → ↑ Cataract grade - ↑ Surgery complexity

#### 3.1 Objective ideas

I had the idea to start selecting the objectives based on the different main phases of the patient's procedure from the beginning to the end:

1. Pre-op: Biometric parameters exam on the patient to do correct calculations for its IOL. Different metrics can be measured for this first surgery part, the identification for the hardness level begins here too. The idea is NOT to create a classification or predictor of cataract levels, already exists. (here will be important the numeric parameters of the patient at the biometric exam, IOL calculations, especially Figure 1)

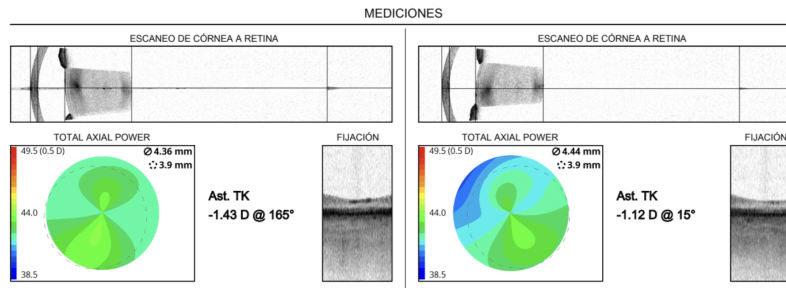


Figure 2: Mediciones encontradas en el archivo WEISS para calculos de IOL.

2. During op: Here's the main development of this project, where we will join the hardness levels with the skills of the surgeon (assessment) having in mind that for each hardness level there's a different amount of skills and experience needed.
3. Post-op: More than using our actual data on this part, it is possible to obtain results for this section by putting more effort during surgery evaluation. This will be the desired output: correct and quick recovery after a successful surgery besides a hard (more surgical ability) or soft (less surgical ability) cataract.

Based on this division, here I have some draft ideas of solutions:

1. Allow the patient to have a better, correct and quick an easy recovery (Post-op)
2. Identification of hand positioning and its change based on cataract hardness (possible to have and identify different levels) to recommend during surgery (During op)
3. Avoid surgery risks during surgery, in general skills, there are some complications that can appear during surgery depending on hardness, tissue and body structure. Risks that can be monitored and how to identify them (*which data can help us with that?*)

**Date of addition: 01/06/2026**

**New additions**

Extracted from Doc Gilberto's infographic → The use of computer vision systems can provide real-time guidance and analysis to improve safety, reduce complications, and enhance surgical outcomes → Many of the common complications are influenced by real-time surgical decisions. Earlier detection (already exists - biometric tests pre-op), better guidance, and objective feedback can significantly reduce risk and improve patient outcomes.

Implementation of CV:

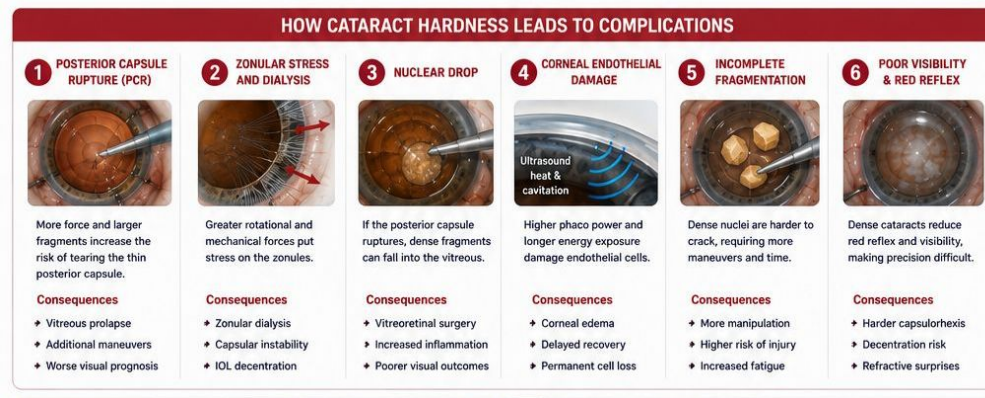
- pre-operative planning
- capsulorhexis guidance
- phacoemulsification safety
- intraoperative risk detection
- IOL implantation assistance
- post-operative analysis

\* Important to note that some of the data can be a limitation on CV implementation, and for that reason, we need to delimit our approach.

These next ideas are based on the last meeting (29/05):

- First implementation: cataract fragments tracking, ID on each fragment to follow them over time via video, and use of color masks
- Special attention to surgical behavior, instrument and tissue trajectory, hand movement, and positioning depending on CATARACT PHASE
- **PROBLEMS: Posterior capsule rupture → its integrity is the main parameter for a successful or unsuccessful cataract surgery**
- Another complication: Zonular dialysis, when the cataract is harder than others, if its pulled with a lot of force, can be ripped
- **Enfoque principal durante el OP, y obtener datos de ayuda recolectados durante el pre-OP en donde implícitamente pueda conseguirse una recuperación exitosa (post-OP), OP → enfoque principal para objetivo principal de skill assessment, supported por parámetros de pre-OP**
- Possibility of a Cataract Coach after the surgery, which can provide a good report

## Doc's Diagram



Based on this segment, it will be possible for us to focus on complications 1 to 4 for skill assessment, which is one of the main topics of the project.

### Idea for one deliverable:

- A system that can define certain scores based on cataract hardness, parameters pre-op, hand positioning, and each phase time. For evaluation, we will define and design protocols and scores for the OP phase (during surgery) → how will the combination of skills and parameters chosen by each surgeon be implemented?